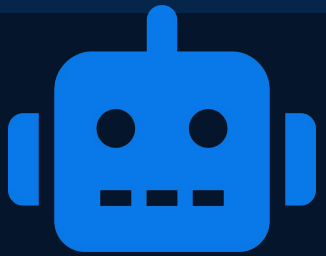
AUTONOMOUS QUANT LAB

Adversarial Alpha Discovery



An Agentic Multi-Agent Framework for Autonomous Scientific Discovery in Quantitative Finance

CREATOR — **CRITIC** — **JUDGE**

Python LangGraph Backtrader ChromaDB yfinance edgartools

THE PROBLEM

Why Traditional Quant Research is Failing

The industry faces four compounding crises that demand an autonomous solution.

Alpha Decay

Profitable inefficiencies disappear once discovered. Competitive markets absorb information faster than humans can generate new edge.

Overfitting

Researchers inadvertently fit strategies to historical noise—models excel in backtests but collapse in live markets.

Look-Ahead Bias

Future information unintentionally leaks into training data, creating unrealistic performance expectations.

Human Bottlenecks

Limited bandwidth, slow hypothesis cycles, and cognitive bias throttle the pace of discovery.

SOLUTION → AI agents autonomously generate hypotheses · Competing agents challenge assumptions · Continuous experimentation at machine speed

Local Data Lake Architecture

S&P 500 · 10 years · OHLCV · yfinance

yfinance API
Source

Data Ingestion
Download & Validate

Local Data Lake
Parquet / SQLite

Backtrader Engine
Simulation-Ready

✓ **Deterministic Testing**
Same dataset every experiment — fully reproducible research.

✓ **No External Dependencies**
No API downtime, latency, or changing data sources.

✓ **Version-Controlled Data**
Enables rigorous benchmarking and scientific repeatability.

✓ **Faster Backtesting**
Local I/O eliminates network overhead.

500 tickers · 10 years · ~1.25M daily bars · OHLCV + Adjusted Close

Extracting Semantic Alpha from SEC Filings

edgartools · 10-K filings · Risk Factors · MD&A

1. **Filing Ingestion**
2. **Text Parsing**
3. **LLM Interpretation**
4. **Economic Signal Extraction**
5. **Strategy Hypothesis**

Risk Factors

→ Operational threats

→ Regulatory concerns

→ Competitive risks

MD&A Section

→ Executive commentary

→ Future outlook

→ Strategic priorities

EXAMPLE INSIGHT

Filing signal: "Supply-chain disruptions and inventory uncertainty"

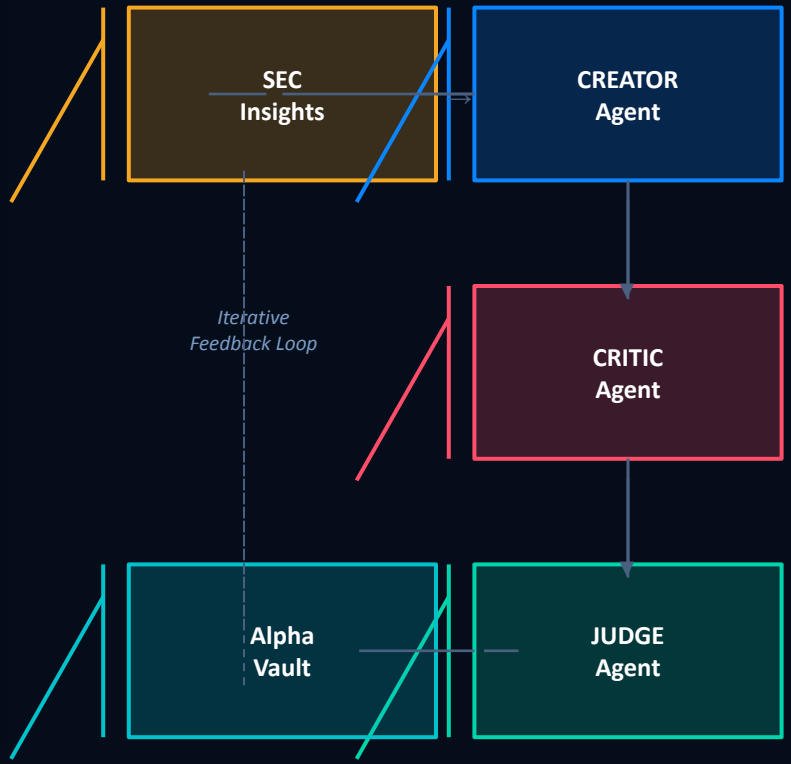
Hypothesis → Firms with elevated supply-chain risk underperform peers during slowdowns.

LangGraph State Machine

Central orchestration engine coordinating all autonomous agents

Research State Object

`sec_insights:`	SEC Insights
`strategy_code:`	Strategy Code
`backtest_metrics:`	Backtest Metrics
`critic_analysis:`	Critic Analysis
`judge_verdict:`	Judge Verdict
`memory_refs:`	Memory References



Secure Strategy Evaluation Environment

Dynamic code execution · Isolated simulation · Standardized metrics

Returns

Sharpe

Win Rate

Drawdown

strategy.py

```python
class AlphaStrategy(bt.Strategy):

    def next(self):

        # Generated by Creator Agent
```

Isolated Execution

Local env · No internet

Fixed Dataset

Controlled inputs only

Repeatable Tests

Consistent benchmarks

AGENT 1 · CREATOR

Alpha Generation Engine

Converts data into executable investment strategies

Economic Interpretation

Transforms SEC insights into market narratives and economic hypotheses.

Strategy Synthesis

Generates entry/exit conditions and risk controls from hypotheses.

Code Production

Outputs fully executable Backtrader strategies as Python code strings.

EXAMPLE

INPUT

Rising regulatory risk across multiple filings.

HYPOTHESIS

Regulatory uncertainty → higher volatility → negative future returns.

STRATEGY

Short high-risk firms · Long low-risk peers

AGENT 2 · CRITIC

The Red Team

A strategy is not robust until it survives adversarial testing.

Transaction Costs

Brokerage commissions on every trade

Slippage

Execution price deviation from signal

Latency

Delayed order execution penalties

Spread Effects

Bid-ask realities on entry & exit

FRAGILITY DETECTION

- ⚠ Overfitting
- ⚠ Data Leakage
- ⚠ Survivorship Bias
- ⚠ Unrealistic Assumptions

STRESS TEST RESULT

2.10

Original Sharpe

→

0.60

After Friction

✗ STRATEGY REJECTED

AGENT 3 · JUDGE

Governance & Memory

Logical consistency · Semantic audit · ChromaDB vector memory

✓ Strategy Validity

Economic justification exists for the hypothesis

✓ Logical Consistency

Strategy behavior matches its stated rationale

✓ Experimental Integrity

Proper testing methodology was followed

ChromaDB Vector Memory

- Previous Hypotheses
- Failed Experiments
- Critic Findings
- Successful Patterns

MEMORY-DRIVEN DECISION

1. Embed new strategy · 2. Retrieve similar history · 3. Analyze prior failures · 4. Issue verdict

Phases 1–3: Building the Foundation

PHASE 01

Data Ingestion Layer

- → Build local market database
- → Download 10 years OHLCV data
- → Collect SEC filings with edgartools

✓ **Structured Local Data Lake**

PHASE 02

Sandbox Engine

- → Configure Backtrader engine
- → Implement dynamic code execution
- → Generate performance reports

✓ **Secure Simulation Environment**

PHASE 03

Vector Ledger

- → Deploy ChromaDB
- → Create experiment memory system
- → Enable semantic retrieval

✓ **Searchable Research History**

Phases 4–6: Autonomous Research

PHASE 04

Creator Integration

→ Connect LLM reasoning layer

→ Automate hypothesis generation

→ SEC → strategy pipeline

✓ **Working Creator Agent**

PHASE 05

Adversarial Loop

→ Build Critic Agent

→ Introduce market friction testing

→ Automate stress evaluation

✓ **Robustness Validation Framework**

PHASE 06

Orchestration

→ Implement LangGraph workflow

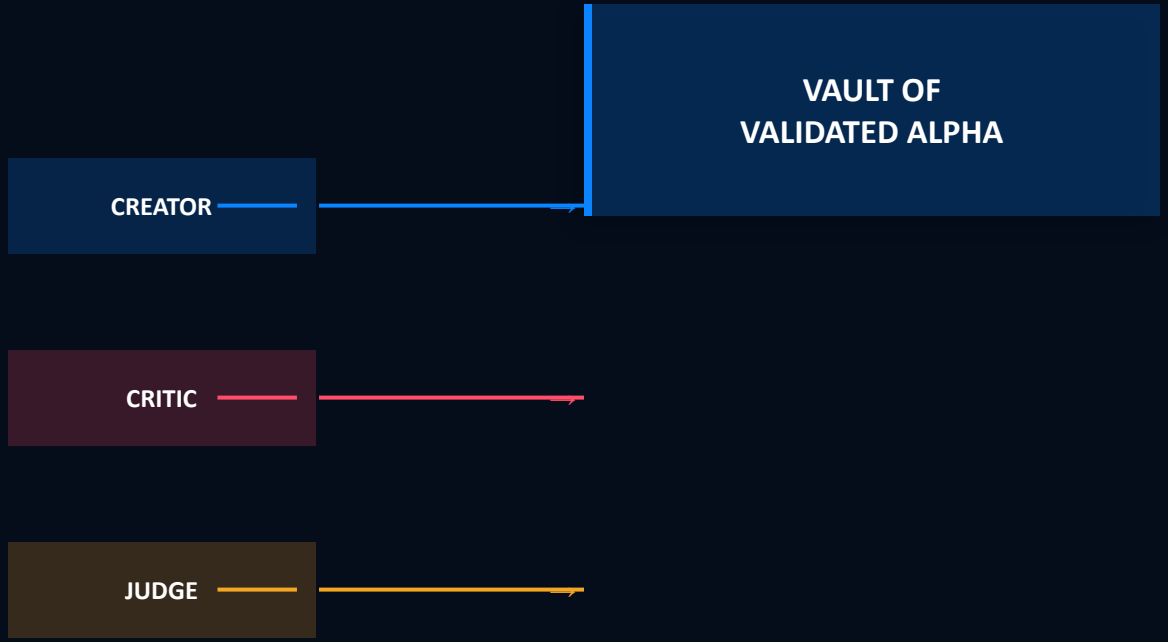
→ Enable end-to-end autonomy

→ Schedule research cycles

✓ **Fully Autonomous Quant Lab**

CONCLUSION

From Research Automation to Alpha Manufacturing



Sharpe Ratio

Risk-adjusted return quality

Max Drawdown

Downside risk exposure

Robustness Score

Friction + stability + survival

An autonomous scientific research system capable of discovering, challenging, and validating investment strategies faster and more rigorously than traditional human-led quantitative research teams.